



Response to the Request for Information Regarding the Temporary Placement of 7-Hydroxymitragynine Above a Specified Threshold in Schedule I

Docket No. HHS-OASH-2026-0232

Executive Summary

The Holistic Alternative Recovery Trust ("HART") was formed in 2023 to provide education to consumers about the safety and efficacy of holistic alternative recovery solutions and to advocate before Congress and state legislatures to promote responsible regulations of plant-based recovery products.

We respectfully submit these comments in response to the Department of Health and Human Services' ("HHS") Request for Information regarding the proposed threshold for the temporary placement of 7-hydroxymitragynine ("7-OH") above a specified threshold in Schedule I of the Controlled Substances Act.

HART supports science-based regulation of botanical products and recognizes the government's legitimate interest in preventing the manufacture and sale of unsafe, adulterated, or deceptively marketed products.

The question presented by this Request for Information, however, is not whether products containing 7-OH should be regulated. Rather, HHS seeks scientific evidence identifying a threshold above which such products present an "imminent hazard to public safety" sufficient to justify temporary placement into Schedule I, which is reserved for those substances classified under the U.S. Controlled Substances Act (CSA) as having:

- High potential for abuse
- No currently accepted medical use in the United States
- Lack of accepted safety under medical supervision

The available scientific evidence does not presently identify such a threshold.

Across pharmacology, toxicology, human clinical experience, and nationwide public-health surveillance, the current record consistently demonstrates that risk associated with orally consumed 7-OH products is determined by multiple interacting factors - including administered dose, oral bioavailability, metabolism, formulation, route of administration, and concomitant substance use - not by concentration alone. (Oral ingestion is the only known route of consumption for 7-OH products.)

Although 7-OH is a pharmacologically active compound and, like other bioactive substances, warrants appropriate regulatory oversight, no published pharmacological, toxicological, epidemiological, or clinical evidence identifies either the proposed 0.050 percent concentration

threshold or the proposed one-milligram-per-article threshold as a biologically meaningful breakpoint associated with a measurable increase in serious adverse outcomes.

That conclusion has been reached independently by experts approaching the issue from different scientific disciplines. A [review](#) of the medical literature and multiple national surveillance systems by Dr. Edward W. Boyer, Professor of Emergency Medicine and Medical Toxicology at The Ohio State University, found no epidemiological signal of overdose mortality or widespread serious toxicity attributable to orally consumed 7-OH. Dr. Kirsten E. Smith, Assistant Professor of Psychiatry and Behavioral Sciences at Johns Hopkins University (2023-2026), [similarly concluded](#) that existing evidence does not demonstrate a significant or unreasonable risk of illness or injury, and instead supports continued research coupled with responsible regulation. [Independent reviews](#) by Drs. Michael Levine, Dr. Jakob Moskovitz, and Dr. Andrew Monte likewise found no evidence that currently available preclinical or surveillance data identify an imminent public-health emergency requiring the proposed numerical thresholds. And Dr. Steven B. Bird [concluded](#) that neither proposed threshold reflects established pharmacological risk and **recommends a finished-product threshold of five (5) mg per dosage unit.**

These scientific conclusions are reinforced by real-world public-health surveillance. Federal monitoring systems - including the National Poison Data System (NPDS), the National Drug Early Warning System (NDEWS), and FDA's Adverse Event Reporting System (FAERS) - exist specifically to detect emerging drug threats. Historically, substances presenting imminent hazards rapidly generate recognizable patterns of poison-center reports, emergency department presentations, published case reports, forensic findings, and overdose surveillance signals. Despite billions of servings of commercially available oral 7-OH products consumed in the United States, no comparable pattern has emerged. The absence of a consistent epidemiological signal across multiple independent surveillance systems is itself scientifically significant.

Accordingly, these comments reach three principal conclusions.

First, the current scientific record does not support either of the proposed numerical thresholds. While the available evidence confirms that 7-OH is pharmacologically active and merits continued study and appropriate regulation, it does not identify either the proposed concentration threshold or the proposed one-milligram threshold as scientifically established points at which products begin presenting an imminent hazard to public safety.

Second, if HHS nevertheless determines that a numerical threshold should be recommended as a matter of regulatory policy, that threshold should be based on actual consumer exposure rather than an easily-gamed standard like percentage concentration. Milligram-per-serving limits more accurately reveal pharmacological risk, are consistent with established toxicological principles, best reflect the way Americans understand consumer product potency, and mirror Congress's recent approach to regulating comparable products.

Finally, any threshold framework should be revised to ensure it is scientifically coherent, administrable, and directed toward products presenting the greatest potential public-health concern. Specifically, DEA should clarify that thresholds apply only to finished consumer products, define "article" as a finished single-serving dosage form, eliminate the percentage-concentration trigger, and increase the finished-product threshold from 1 mg to 5 mg per dosage unit. These revisions would preserve DEA's ability to address genuinely high-potency

products while better aligning the temporary scheduling framework with the available scientific evidence and modern regulatory practice.

I. The Scientific Record Does Not Support the Proposed Thresholds

The Request for Information poses a straightforward scientific question: does the available evidence identify a concentration or dosage threshold above which products containing 7-hydroxymitragynine ("7-OH") present an imminent hazard to the public safety?

The current scientific record does not answer that question in the affirmative.

The available evidence demonstrates that 7-OH is a pharmacologically active compound that warrants continued scientific study and appropriate regulatory oversight. However, it does not identify either the proposed 0.050 percent concentration threshold or the proposed one-milligram-per-article threshold as scientifically established breakpoints associated with a measurable increase in serious adverse outcomes. Rather, the evidence consistently demonstrates that risk is determined by multiple interacting pharmacological, toxicological, and clinical variables that cannot be reduced to concentration alone.

Importantly, this conclusion is not derived from a single study, researcher, or even discipline. Independent experts in pharmacology, medical toxicology, emergency medicine, behavioral science, poison surveillance, and clinical research - including Drs. Steven Bird, Edward Boyer, Kirsten Smith, Michael Levine, Jakob Moskovitz, and Andrew Monte - reviewed different portions of the available evidence and nevertheless reached remarkably consistent conclusions. Although they approached the issue from the perspective of different scientific disciplines, none identified evidence supporting DEA's proposed numerical thresholds as biologically meaningful indicators of imminent public-health risk.

A. Pharmacology Does Not Identify a Scientifically Meaningful Breakpoint

The fact that 7-OH exhibits pharmacological activity does not itself establish an appropriate regulatory threshold. Modern pharmacology distinguishes among [pharmacological potency](#), [systemic exposure](#), and [actual clinical risk](#). While 7-OH functions as a partial agonist at the μ -opioid receptor, receptor binding alone does not predict toxicity. Instead, clinical effects depend upon the amount of drug that ultimately reaches systemic circulation and target tissues following oral administration.

The pharmacology of 7-OH further illustrates why receptor affinity alone is an incomplete measure of public-health risk. Although 7-OH binds with high affinity to the μ -opioid receptor, it is a G protein-biased partial agonist that exhibits relatively low intrinsic efficacy, more closely resembling buprenorphine than full μ -opioid agonists. ([Kruegel](#), 2016; [Obeng](#), 2021; [Todd](#), 2020; [Varadi](#), 2016; [WHO](#), 2021). As a partial agonist, 7-OH does not fully activate the receptor even when receptor occupancy increases, producing a pharmacological ceiling on opioid-like effects that distinguishes it from conventional opioids.

This distinction is particularly important because public discussions of 7-OH frequently conflate potency with intrinsic efficacy. While 7-OH demonstrates higher receptor potency than morphine in [certain laboratory assays](#) involving guinea pigs (and not humans), potency [reflects](#) the amount of drug required to produce a given effect—not the maximum effect the drug is capable of producing. Because 7-OH is a [partial agonist with comparatively low intrinsic efficacy](#),

statements that it is "more potent than morphine" do not establish that it produces morphine-like toxicity or overdose risk in humans.

Consistent with this pharmacological profile, experimental studies further demonstrate that 7-OH produces [substantially reduced \$\beta\$ -arrestin-2 recruitment](#) compared with conventional opioid agonists. This signaling pathway has been associated with respiratory depression and other dose-limiting adverse effects of classical opioids. ([Todd](#), 2020; [Samways](#), 2024). More importantly, federally funded research conducted at Memorial Sloan Kettering Cancer Center identified that 7-OH does not induce respiratory depression (DoD award number W81XWH-17-1-0256, "Safer nonaddicting and nonabusable analgesics....," Majumda).

While β -arrestin signaling alone cannot predict clinical outcomes, these findings provide a biologically plausible explanation for why the available preclinical, clinical, and epidemiological evidence has not demonstrated the pattern of severe respiratory toxicity typically associated with full μ -opioid agonists.

These mechanistic differences are further amplified by 7-OH's pharmacokinetics. Orally administered 7-OH exhibits extremely poor oral bioavailability - approximately 2.7 percent - meaning that only a small fraction of an ingested dose survives first-pass metabolism and reaches systemic circulation. ([Chiang et al.](#), 2025). Accordingly, receptor potency analyses tend to substantially overstate exposure experienced following oral consumption of commercially available products.

As Dr. Steven B. Bird explains in his [independent toxicological review](#), a finished dosage unit containing up to 5 mg of 7-OH would be expected to produce plasma concentrations approximately fifteen times lower than those observed in published human kratom studies that reported no signal of serious toxicity or respiratory depression. These findings undergird Dr. Bird's recommendation of a 5mg dosage limit per article, and the aforementioned pharmacokinetic characteristics substantially limit systemic exposure and refute the scientific premise that either the proposed concentration threshold or the proposed one-milligram limit corresponds to a meaningful increase in pharmacological risk.

Dr. Edward Boyer's [2025 review](#) reached the same conclusion from a different perspective. Evaluating FDA's 2025 scientific assessment, Dr. Boyer explained that the agency places undue emphasis on receptor affinity while giving insufficient consideration to exposure-response relationships – the very principle that FDA has long recognized as central to evaluating drug safety. As FDA's own [guidance](#) explains, "exposure-response information is at the heart of any determination of safety." In Dr. Boyer's view, pharmacological potency, standing alone, is an inadequate predictor of toxicity because it fails to account for the pharmacokinetic factors that determine actual human exposure.

Dr. Boyer noted that the compound exhibits poor oral bioavailability, short residence time within the body, and limited penetration into the central nervous system. Collectively, these characteristics substantially reduce systemic exposure following oral consumption and help explain why the expected epidemiological signal associated with highly toxic opioid agonists has not emerged despite widespread consumer use.

These principles carry important regulatory implications. Products containing identical concentrations of 7-OH may produce materially different systemic exposures depending upon

formulation, dosage form, absorption characteristics, metabolism, and route of administration. Conversely, products with substantially different concentration percentages may expose consumers to similar amounts of active compound. Put simply, concentration is not synonymous with exposure, and exposure – not nominal concentration – is the relevant determinant of pharmacological risk.

For this reason, neither Dr. Bird nor Dr. Boyer identifies the proposed 0.050 percent concentration threshold or the proposed one-milligram-per-article threshold as scientifically meaningful pharmacological breakpoints. Instead, both conclude that actual consumer exposure provides the more scientifically appropriate framework for evaluating risk.

Finally, and perhaps most consequentially, the relative innocuousness of 7-OH was seemingly confirmed by FDA-affiliated researchers, who led a [2026 randomized control trial](#) on kratom in which no serious adverse events (SAEs) occurred among human subjects. Investigators administered increasing doses of kratom powder at levels of 1.5% mitragynine, up to 12,000mg (12g). Since the human liver [converts approximately 25% of mitragynine into 7-OH](#), study participants received up to 45mg of 7-OH ($12,000 * .015 * .25$ – a higher dosage than most consumers use, according to industry surveys. Quoting directly from the recently-published study: “No deaths or serious adverse events occurred.”

Accordingly, the available pharmacological evidence does not establish that products abruptly transition from presenting ordinary pharmacological activity to constituting an imminent hazard to public safety at either of DEA's proposed numerical thresholds. Rather, the evidence demonstrates that pharmacological risk exists along a continuum governed by exposure, dose, formulation, metabolism, and route of administration. To the extent HHS ultimately recommends a numerical threshold as a matter of regulatory policy, that threshold should therefore be understood as a policy judgment informed by science – not a value uniquely dictated by the current scientific record.

B. Preclinical Toxicology Does Not Identify an Imminent Hazard or Support the Proposed Thresholds

The available preclinical toxicology likewise does not identify a scientifically meaningful concentration or dosage threshold supporting DEA's proposed framework.

Animal and in vitro studies are an essential component of safety assessment because they help identify potential toxicological concerns before widespread human exposure occurs. At the same time, toxicology has long recognized that preclinical findings must ultimately be interpreted alongside pharmacokinetics, clinical experience, and real-world epidemiological data. Animal models are designed to [identify potential hazards](#) - not to establish the precise regulatory thresholds at which products present an imminent public-health emergency *for humans*.

The currently available preclinical evidence regarding orally administered 7-OH does not demonstrate the type of toxicological profile ordinarily associated with emergency regulatory intervention. [Independent reviews](#) conducted by Dr. Michael Levine, Professor of Emergency Medicine and Co-Chief of the Division of Medical Toxicology at UCLA, and Dr. Jakob Moskovitz, Associate Professor of Pharmacology and Toxicology at the University of Kansas School of Pharmacy, reached substantially similar conclusions despite evaluating different aspects of the available evidence. Both concluded that existing preclinical studies support

continued scientific investigation but do not identify evidence of serious toxicity sufficient to characterize currently marketed oral 7-OH products as presenting an imminent public-health hazard.

Dr. Levine [reviewed](#) both a rodent toxicity study and a canine safety study, focusing specifically on the types of adverse effects that would ordinarily be expected from a compound acting at the μ -opioid receptor - particularly central nervous system depression and respiratory depression. His review found that no animals exhibited serious clinical toxicity during the course of either study. While minor findings such as transient drooling and gastrointestinal abnormalities were observed in some animals, similar findings also occurred in control animals, making a causal relationship to 7-OH difficult to establish. Dr. Levine ultimately concluded that the available preclinical evidence demonstrated no serious toxicological signal and supported further investigation through appropriately designed human studies rather than suggesting an imminent safety concern.

Dr. Moskovitz independently [evaluated](#) the available in vitro, rodent, and canine studies and reached remarkably similar conclusions. His review found that 7-OH demonstrated no clinically significant evidence of cardiac, hepatic, or renal toxicity in cell-based assays and that no meaningful in-life toxicity was observed in the animal models. Indeed, he concluded that the available evidence suggests that the safety profile of 7-OH is comparable to - or in certain respects more favorable than - that of the related naturally occurring kratom alkaloid mitragynine, which will remain legal under this scheduling order. Based upon the totality of the available laboratory and animal evidence, Dr. Moskovitz concluded that the data do not support the conclusion that 7-OH presents a significant or serious safety concern.

These conclusions are notable not because they establish that 7-OH is entirely without risk – no pharmacologically active compound is – but because they demonstrate what the current evidence does not show: a toxicological breakpoint corresponding to either DEA's proposed 0.050 percent concentration threshold or its proposed one-milligram-per-article threshold. Nor does it demonstrate the severe respiratory depression, persistent neurological impairment, target-organ toxicity, or other serious adverse effects that ordinarily precede emergency scheduling actions involving newly emerging substances.

Dr. Steven Bird likewise [observed](#) that the available pharmacological and toxicological data do not support the proposition that products containing up to 5 mg of orally administered 7-OH present an imminent public-health hazard. Instead, his analysis concludes that a 5 mg finished-product threshold remains conservative when considered alongside the compound's limited oral bioavailability and the absence of corresponding evidence of serious toxicity in the available preclinical literature.

Taken together, the available animal, in vitro, and pharmacokinetic evidence supports a consistent conclusion. The existing preclinical record identifies the need for continued research - as would be expected for any relatively new bioactive compound - but it does not identify either of DEA's proposed numerical thresholds as scientifically established toxicological breakpoints. Instead, the evidence supports continued investigation, appropriate manufacturing controls, and ongoing surveillance while human clinical data continue to develop.

C. Human Experience Does Not Demonstrate Widespread Serious Harm or Identify a Scientifically Meaningful Threshold

Ultimately, the most probative evidence of public-health risk is real-world human experience. While pharmacological and preclinical studies provide important information regarding biological activity and potential toxicity, they cannot substitute for evidence demonstrating how a substance actually performs following widespread human use. Here, the available human evidence weighs strongly against the conclusion that 7-OH products present an imminent hazard to public safety or that either of DEA's proposed numerical thresholds corresponds to a measurable increase in serious adverse outcomes.

This conclusion is particularly significant because commercially available 7-OH products have now been consumed on an extraordinary scale. By conservative estimates, more than [one million Americans](#) have consumed billions of servings of these products, nearly all in dosages vastly exceeding the proposed 1 mg threshold. Had currently marketed oral 7-OH products been producing the type of severe toxicity ordinarily associated with an emerging public-health crisis, one would reasonably expect to observe a corresponding increase in published case reports, poison-center alerts, emergency department presentations, overdose investigations, and forensic findings. That pattern has not emerged.

To HART's knowledge, after reviewing published case reports, FDA adverse-event reports, poison-center data, and the expert analyses submitted in this proceeding, there is no confirmed fatality in which orally consumed 7-OH ingested in isolation has been established as the cause of death - despite billions of servings consumed. Fatalities reported in surveillance systems almost [invariably involve](#) significant polysubstance exposure (often alongside heroin, fentanyl, or other substances known to cause death), or other substantial confounding factors, and several cases cited in public discussions predate the commercial introduction of modern 7-OH products altogether. Although the absence of confirmed isolated fatalities does not by itself establish safety, it is highly probative when considered alongside the compound's pharmacology, the available toxicological evidence, and the absence of a broader epidemiological signal.

Dr. Edward W. Boyer's [comprehensive review](#) of the published literature, poison-center surveillance, adverse-event reports, forensic data, and federal monitoring systems found [no published case reports](#) documenting acute poisoning or overdose deaths directly attributable to 7-OH products.

Medical toxicologists have historically identified emerging public-health threats with remarkable speed, including outbreaks involving synthetic cannabinoids and tianeptine. According to Dr. Boyer, when a substance is producing significant overdose morbidity or mortality, multiple cases rapidly generate recognizable epidemiological signals. In the case of 7-OH, however, no such signal has emerged.

Dr. Boyer further [concluded](#) that the absence of overdose reports is not simply the product of underreporting or unfamiliarity with the compound. Rather, it reflects the fact that nationwide surveillance systems specifically designed to detect emerging toxicological threats have not identified evidence of widespread serious harm associated with 7-OH. As he explains, "multiple overdoses generate signals," yet no comparable signal has appeared despite years of increasing consumer exposure. This observation is particularly significant because the same surveillance infrastructure has successfully identified numerous emerging substances posing genuine public-health concerns long before they became widely recognized.

Dr. Kirsten E. Smith, one of the nation's leading researchers on kratom and its alkaloids, independently reached [substantially similar conclusions](#) after reviewing preclinical evidence, clinical literature, poison-center data, adverse-event reporting systems, and her own ongoing research involving U.S. consumers. Her assessment is particularly valuable because it combines traditional toxicological evidence with direct observational research involving individuals who regularly consume 7-OH products.

Dr. Smith concluded that existing evidence is inadequate to support the claim that 7-OH products present a significant or unreasonable risk of illness or injury. While acknowledging that physical dependence and substance use disorder are possible - as they are with many pharmacologically active substances - she [found no evidence](#) that severe illness, widespread addiction, or population-level harm has emerged among current consumers. Her research, including surveys and qualitative interviews with individuals using commercially available 7-OH products, instead suggests that many consumers report meaningful therapeutic or quality-of-life benefits while experiencing few serious adverse effects.

Importantly, Dr. Smith also evaluated one of the principal concerns often raised regarding compounds acting at the μ -opioid receptor: respiratory depression. After reviewing both the preclinical and clinical evidence, she concluded that currently available human data [do not demonstrate](#) widespread or clinically significant respiratory depression associated with orally consumed 7-OH. Although very high experimental doses in animal models – administered, in at least one case, intravenously – have produced transient respiratory effects, these findings have not translated into a recognizable pattern of respiratory failure or fatal overdose among human consumers using commercially available oral products.

Neither expert suggests that 7-OH is entirely without risk. To the contrary, both acknowledge that continued scientific investigation remains appropriate and that products containing 7-OH should be manufactured, labeled, and marketed responsibly. Their conclusions instead reflect a more limited – but critically important – scientific proposition: the currently available human evidence does not identify either DEA's proposed 0.050 percent concentration threshold or its proposed one-milligram-per-article threshold as scientifically meaningful points at which products begin presenting an imminent hazard to public safety.

This distinction is critical because the question before HHS is not whether products containing 7-OH should be subject to appropriate regulatory safeguards. Rather, it is whether the available scientific evidence supports identifying a specific numerical threshold that poses an imminent threat to public health. At this time, the published literature, clinical experience, adverse-event data, and observational research do not support that conclusion.

Accordingly, the existing human evidence supports continued surveillance, additional clinical research, appropriate manufacturing standards, and responsible product regulation. It does not support the conclusion that either of the proposed numerical thresholds is uniquely established by the current scientific record.

D. Public Health Surveillance Does Not Demonstrate an Emerging Public Health Emergency

Independent public-health surveillance provides perhaps the strongest test of whether a substance presents an imminent hazard to public safety. Unlike preclinical research or isolated case reports,

surveillance systems evaluate population-level outcomes across millions of real-world exposures. They are specifically designed to identify emerging patterns of overdose, poisoning, hospitalization, and death that may warrant rapid public-health or regulatory intervention.

Historically, substances posing genuine and immediate public-health threats generate recognizable epidemiological signals shortly after widespread availability. Synthetic cannabinoids, synthetic cathinones, fentanyl analogues, and tianeptine all produced significant increases in poison-center calls, emergency department presentations, forensic toxicology findings, published case reports, and overdose fatalities that were rapidly detected through national surveillance systems. Those signals formed an important part of the scientific basis for subsequent regulatory action.

The available evidence concerning commercially available 7-OH products presents a markedly different picture.

As Dr. Edward Boyer [explained](#), the United States maintains multiple complementary surveillance systems capable of identifying emerging toxicological threats, including the National Poison Data System ("NPDS"), DEA TOX, the National Drug Early Warning System ("NDEWS"), FDA's Adverse Event Reporting System ("FAERS"), published medical case reports, and forensic toxicology reporting. Collectively, these systems monitor poison-center consultations, emergency department encounters, toxicological investigations, adverse-event reporting, overdose deaths, and trends in illicit drug markets. According to Dr. Boyer's review, none of these independent surveillance systems has identified a pattern of widespread serious toxicity attributable to 7-OH products.

This absence of a consistent epidemiological signal is scientifically significant. Surveillance systems are inherently imperfect, and isolated adverse events may go unreported. However, the purpose of these systems is not to capture every individual incident; rather, they are designed to detect meaningful population-level trends. When a substance produces substantial morbidity or mortality across large numbers of users, those trends ordinarily become visible across multiple independent datasets. As Dr. Boyer [observed](#), there is a "lack of [a] public health signal of adverse effects from 7OH." The failure of multiple surveillance systems to identify such a signal despite extensive consumer exposure weighs against the conclusion that currently marketed oral 7-OH products present an imminent public-health emergency.

Dr. Kirsten Smith independently reviewed many of the same surveillance datasets and reached [substantially similar conclusions](#). Her assessment found no evidence of widespread illness, injury, or overdose sufficient to support the conclusion that oral 7-OH products present a significant or unreasonable risk to the public. Although adverse events have been [reported](#) - as would be expected with virtually any pharmacologically active product - the available data do not reveal the type of consistent, reproducible pattern ordinarily associated with an emerging national health crisis. Instead, the existing evidence supports continued monitoring while additional clinical and epidemiological data are developed.

Dr. Andrew Monte, Professor of Emergency Medicine and Medical Toxicology at the University of Colorado School of Medicine and a principal investigator with the RADARS® System, [likewise concluded](#) that the available surveillance evidence does not suggest an unreasonable risk to public health. Dr. Monte's review is particularly informative because RADARS® was specifically developed to identify emerging patterns of misuse, abuse, diversion, overdose, and

other adverse outcomes associated with prescription and non-prescription substances. Based on the currently available evidence, Dr. Monte concluded that continued surveillance remains appropriate but that existing data do not support the conclusion that commercially available 7-OH products are producing a level of harm warranting the proposed numerical thresholds.

These conclusions are further reinforced by the scale of consumer exposure. According to [independent market analyses](#), more than one million Americans have consumed, per industry estimates, billions of commercially available 7-OH products since their introduction to the marketplace in 2023. While widespread utilization alone cannot establish safety, it substantially increases the likelihood that meaningful safety concerns would become apparent through the very surveillance systems designed to detect them. The absence of a consistent signal across multiple independent surveillance programs after such extensive exposure is therefore an important component of the overall scientific record.

Importantly, this does not mean that surveillance should cease or that regulation is unnecessary. On the contrary, continued monitoring through NPDS, NDEWS, DEA TOX, RADARS®, FAERS, and other established public-health systems remains essential as consumer use evolves and additional research becomes available. Surveillance is most valuable when it informs evidence-based regulatory decision-making over time rather than being viewed as a one-time exercise.

Taken together, the pharmacological evidence, preclinical toxicology, human clinical experience, and nationwide surveillance data lead to the same conclusion. The current scientific record supports continued research, ongoing surveillance, responsible manufacturing standards, and appropriate regulatory oversight. It does not identify either the proposed 0.050 percent concentration threshold or the proposed one-milligram-per-article threshold as scientifically established points at which products begin presenting an imminent hazard to public safety.

Accordingly, if HHS determines that numerical thresholds remain appropriate as a matter of regulatory policy, those thresholds should be understood as policy judgments informed by the available science - not as values uniquely dictated by the current scientific record. The following recommendations in Part III of this response are offered with that distinction in mind.

II. Emergency Scheduling Should Be Reserved for Demonstrable Public Health Emergencies

Congress vested the [temporary scheduling authority](#) in the Controlled Substances Act to enable the federal government to respond rapidly to extraordinary and emergent threats to public health. By design, it is an exceptional regulatory mechanism that permits expedited action before the completion of the ordinary rulemaking process. That authority is therefore most appropriately reserved for circumstances in which the available scientific evidence demonstrates an imminent hazard requiring immediate intervention.

The scientific record summarized above does not present that circumstance.

The evidence confirms that 7-OH is a pharmacologically active compound that warrants continued study and appropriate regulatory oversight. However, across pharmacology, toxicology, human experience, and nationwide surveillance, the current record does not identify

the hallmarks of an emerging public-health emergency. Independent experts reviewing different aspects of the available evidence consistently conclude that the proposed numerical thresholds are not scientifically established breakpoints associated with a measurable increase in serious harm. Likewise, the national surveillance systems specifically designed to detect emerging drug threats have not identified the type of consistent epidemiological signal that has historically preceded emergency regulatory action against newly emerging substances.

The absence of evidence supporting emergency scheduling does not imply that regulation is unnecessary. Agencies routinely regulate products presenting varying degrees of risk through ordinary administrative processes informed by evolving scientific evidence. Temporary scheduling, however, occupies a different place within the statutory framework. Because it authorizes immediate and sweeping regulatory consequences, it should be reserved for situations in which the scientific record demonstrates an imminent threat that cannot reasonably await the ordinary deliberative process.

Nothing in these comments should be understood as discouraging HHS from recommending appropriate safeguards for products containing 7-OH. To the contrary, HART supports reasonable manufacturing standards, age restrictions, product testing, labeling requirements, and continued surveillance to ensure consumer safety. The question presented by this Request for Information, however, is considerably narrower: whether the current scientific evidence supports the extraordinary remedy of temporary Schedule I placement through DEA's emergency scheduling authority.

Based on the record presently before the Agency, it does not.

Accordingly, if HHS concludes that additional federal controls are warranted, it should recommend that DEA proceed through the ordinary scheduling process established by the Controlled Substances Act. That process is specifically designed to permit comprehensive scientific evaluation, robust public participation, and reasoned administrative decision-making before imposing the substantial legal consequences associated with permanent scheduling. By contrast, temporary scheduling is an extraordinary authority intended to address imminent hazards to public safety. The current scientific record does not demonstrate the type of immediate public-health emergency that justifies bypassing the ordinary scheduling process Congress established.

III. Improving the Proposed Threshold Framework

Although the scientific record discussed above does not support either the proposed 0.050 percent concentration threshold or the proposed one-milligram-per-article threshold as scientifically established indicators of an imminent hazard to public safety, HART recognizes that HHS and DEA may conclude, as a matter of regulatory policy, that numerical thresholds are appropriate to distinguish between products presenting materially different levels of potential risk.

If so, those thresholds should be structured in a manner that reflects established toxicological principles, can be implemented consistently by regulated manufacturers, and directs enforcement resources toward products presenting the greatest potential public-health concern. Several

targeted revisions would substantially improve the proposed framework while remaining faithful to the scientific record.

A. Clarify That Thresholds Apply Only to Finished Consumer Products

The proposed rule repeatedly refers to an "article" containing more than one milligram of 7-OH but does not define that term with sufficient precision. This ambiguity creates unnecessary uncertainty for manufacturers, laboratories, distributors, and enforcement personnel.

For purposes of temporary scheduling, DEA should expressly define an "article" as a finished consumer dosage unit intended for retail sale or consumption, such as a tablet, capsule, beverage, gummy, or other single-serving product.

Applying thresholds to intermediate manufacturing materials, bulk ingredients, or works-in-progress would create significant compliance difficulties without advancing the Agency's stated public-health objectives. Concentrated intermediate materials are routinely used during lawful manufacturing processes before dilution into finished products. Regulating those materials under a finished-product threshold could inadvertently criminalize ordinary manufacturing practices that never result in consumer exposure.

Limiting the threshold to finished consumer products would more accurately align regulatory oversight with actual public health risk while providing regulated entities with a clear and administrable compliance standard.

B. Eliminate the Percentage Concentration Threshold

Among the two proposed numerical thresholds, the percentage concentration limit is the least scientifically defensible.

As discussed in Part I, pharmacological risk is determined by **systemic exposure**, not concentration expressed as a percentage of product weight. Two products containing identical concentration percentages may produce dramatically different consumer exposures depending upon serving size, dosage form, oral bioavailability, and formulation. Conversely, products with substantially different concentration percentages may deliver nearly identical amounts of active compound.

Dr. Steven Bird specifically [concludes](#) that concentration-based regulation does not accurately reflect pharmacological risk because concentration alone fails to measure the quantity of 7-OH actually consumed. Dr. Edward Boyer likewise emphasizes that toxicological risk depends upon exposure-response relationships rather than receptor affinity or nominal concentration.

These conclusions are consistent with longstanding toxicological principles. Hazard is not determined simply by the proportion of active ingredient within a product, but by the amount ultimately absorbed by the consumer under real-world conditions of use.

For these reasons, HHS should recommend that DEA eliminate the proposed 0.050 percent concentration threshold and instead regulate finished products using a single exposure-based standard.

C. Increase the Finished-Product Threshold to 5 mg Per Dosage Unit

If HHS concludes that a finished-product threshold remains appropriate, the scientific record supports increasing that threshold from one milligram to five milligrams per finished dosage unit.

Dr. Steven Bird's pharmacokinetic [analysis](#) demonstrates that a five-milligram oral dosage would still produce systemic exposure substantially below plasma concentrations observed in published human kratom studies that reported no serious toxicity. Because oral bioavailability remains approximately 2.7 percent, only a small fraction of an ingested dose reaches systemic circulation.

Importantly, no published pharmacological, toxicological, clinical, or epidemiological evidence identifies one milligram as a biologically meaningful threshold. Rather, the proposed limit appears to represent a conservative judgment rather than a value derived from experimentally observed changes in human risk.

A five-milligram threshold would therefore remain protective while better reflecting the available scientific evidence. It would also distinguish lower-exposure finished products from significantly higher-potency formulations without unnecessarily encompassing products for which the current record does not demonstrate an imminent public-health hazard.

Such an approach appropriately balances consumer safety, regulatory clarity, and scientific integrity.

D. Continue Surveillance and Encourage Additional Research

Finally, HHS should recommend that DEA continue to evaluate emerging evidence as additional clinical and epidemiological data become available.

The experts whose analyses are discussed throughout these comments consistently recognize that the scientific literature concerning 7-OH continues to evolve. Continued monitoring through NPDS, DEA TOX, NDEWS, RADARS®, FAERS, and other established surveillance systems will provide regulators with increasingly robust information regarding patterns of use, adverse events, misuse, and long-term outcomes.

Similarly, additional human pharmacokinetic studies, controlled clinical investigations, and epidemiological research will further refine understanding of dose-response relationships and product-specific risk.

Regulatory frameworks governing emerging substances should evolve alongside the evidence. Continued surveillance and research will allow HHS and DEA to reassess future policy decisions based upon increasingly comprehensive scientific data while ensuring that regulatory actions remain proportional to demonstrated public-health risk.

IV. Conclusion

HHS requested scientific evidence to inform whether products containing 7-hydroxymitragynine above specified thresholds should be recommended for temporary placement in Schedule I. The available record provides an increasingly robust understanding of the pharmacology and toxicology of 7-OH. It does not, however, identify either the proposed 0.050 percent concentration threshold or the proposed one-milligram-per-article threshold as scientifically established indicators of an imminent hazard to public safety.

Across multiple independent disciplines, the evidence points to the same conclusion. Pharmacological research demonstrates that systemic exposure - not concentration alone - determines risk. Preclinical toxicology has not identified severe toxicity corresponding to the proposed numerical thresholds. Human clinical experience has not demonstrated widespread serious harm associated with commercially available oral products. National surveillance systems designed specifically to detect emerging drug threats likewise have not identified the epidemiological patterns ordinarily associated with substances presenting an imminent public-health emergency.

Importantly, these conclusions are not based upon a single study or individual expert. Independent reviews conducted by Drs. Steven Bird, Edward Boyer, Kirsten Smith, Michael Levine, Jakob Moskovitz, and Andrew Monte – approaching the issue from pharmacological, toxicological, emergency medical, psychiatric, and public-health surveillance perspectives – consistently conclude that the current scientific record does not establish the proposed thresholds as biologically meaningful indicators of risk. While each recognizes the importance of continued research and appropriate regulatory oversight, none concludes that the available evidence supports the specific numerical limits proposed by DEA.

HART likewise recognizes that products containing pharmacologically active compounds should be manufactured responsibly and regulated appropriately. These comments should not be interpreted as opposing reasonable safeguards designed to protect consumers or ensure product quality. Rather, they urge HHS to ensure that any recommendation to DEA accurately reflects what the science presently demonstrates - and, equally important, what it does not.

Accordingly, HART respectfully recommends that HHS:

1. **Conclude that the current scientific record does not support either the proposed 0.050 percent concentration threshold or the proposed one-milligram-per-article threshold as scientifically established indicators of an imminent hazard to public safety;**
2. **Recommend elimination of the proposed percentage-concentration threshold in favor of a finished-product standard based upon actual consumer exposure;**
3. **Recommend increasing the finished-product threshold to 5 mg of 7-OH per dosage unit, consistent with Dr. Steven Bird's pharmacokinetic analysis;**
4. **Clarify that any threshold applies only to finished consumer dosage units; and**
5. **Encourage continued clinical research and public-health surveillance so that future regulatory decisions may be informed by an increasingly comprehensive scientific record.**

The Controlled Substances Act entrusts HHS with an essential scientific advisory role. That role is not simply to identify potential hazards, but to distinguish between what has been demonstrated by evidence and what remains uncertain. Here, the evidence supports continued study, continued surveillance, and appropriately tailored regulation. It does not establish that DEA's proposed numerical thresholds are themselves grounded in the current scientific record.

For these reasons, HART respectfully urges HHS to recommend that DEA hold its temporary scheduling recommendation in abeyance and revise the proposed framework in a manner that more accurately reflects the available scientific evidence.

Additionally, HART respectfully urges HHS to recommend that any further evaluation of 7-OH proceed through the Act's ordinary scheduling process, where the scientific evidence can

continue to develop and the Agency can build a complete administrative record before determining whether permanent scheduling is warranted.

Sincerely,

Jeff Smith, Ph.D

Executive Director

Holistic Alternative Recovery Trust

jeff@hartsupporter.com

Sources Cited

I. Independent Experts

Dr. Steven B. Bird, MD, FACMT, FACEP, is a nationally recognized emergency physician, board-certified medical toxicologist, and Professor of Emergency Medicine at UMass Chan Medical School, where he serves as Division Director of Medical Toxicology. A graduate of Yale University and Northwestern University Feinberg School of Medicine, Dr. Bird completed his emergency medicine residency and medical toxicology fellowship at UMass following service as a U.S. Navy Flight Surgeon with the Marine Corps. He is a past president of the Society for Academic Emergency Medicine and has authored more than 90 peer-reviewed publications on toxicology, poisoning, pharmacology, and emergency medicine. Widely regarded as one of the nation's leading experts in clinical toxicology and the health effects of drugs and chemical exposures, Dr. Bird has frequently served as an expert witness in complex litigation and regulatory matters, translating scientific evidence into practical public health and medical policy.

Edward W. Boyer, M.D., Ph.D. Dr. Boyer is a board-certified emergency physician and medical toxicologist with over 25 years of clinical experience in treating adverse drug events, toxicities, and substance use disorders. He is a Professor with Tenure at The Ohio State University and serves on the staff of the Central Ohio Poison Control Center. He has published extensively on kratom and its alkaloids, has served as an NIH-funded researcher, and is recognized for his expertise in opioid pharmacology, medical toxicology, and public health surveillance of emerging substances.

Michael Levine, M.D. Dr. Levine is a Professor of Emergency Medicine and Co- chief of the Division of Medical Toxicology at the University of California, Los Angeles (“UCLA”). He completed his medical toxicology fellowship at Banner University Medical Center and has held leadership roles at both the University of Southern California and UCLA. He has authored over 130 peer-reviewed articles on drug toxicity, has served on multiple hospital and academic committees related to medication safety, and is an experienced clinical and preclinical researcher in the field of toxicology.

Andrew A. Monte, M.D., Ph.D. Dr. Monte is a board-certified physician and pharmacologist with dual doctoral degrees in medicine and pharmacology. Dr. Monte is currently a Professor of Pharmacology and Medicine at a leading academic medical center, where his research focuses on the pharmacodynamics and toxicology of opioid and non-opioid analgesics, including kratom alkaloids. He has authored numerous peer-reviewed publications on the safety and efficacy of emerging psychoactive substances and has served as a consultant to federal agencies on issues related to drug safety, abuse potential, and regulatory policy. His expertise encompasses both preclinical and clinical evaluation of novel compounds, and he is widely recognized for his contributions to the field of substance use research.

Jackob Moskovitz, Ph.D. Dr. Moskovitz is an Associate Professor with tenure in the Department of Pharmacology and Toxicology at the University of Kansas School of Pharmacy. He earned his Ph.D. in Medical Sciences from the Technion-Israel Institute of Technology and has held research fellowships at the National Institutes of Health (“NIH”) and Roche Institute. His research focuses on neuropharmacology, oxidative stress, and drug safety, and he has

published extensively in peer-reviewed journals, served as a grant reviewer for major funding agencies, and holds several patents in biomedical research.

Kirsten E. Smith, Ph.D. Dr. Smith served as Assistant Professor in the Department of Psychiatry and Behavioral Sciences at the Johns Hopkins University School of Medicine, specializing in behavioral pharmacology and substance use research. She holds a Ph.D. in social work with a focus on drug use and has extensive experience conducting clinical and survey research on kratom, MG, 7-OH, and related compounds. She is a recognized expert in kratom science, has published widely in peer-reviewed journals, and led the Kratom Research Unit at Johns Hopkins, with multiple NIH-funded projects and advisory roles for public health and regulatory bodies.

7-Hydroxymitragynine Science Update

Introduction: Kratom is a plant native to Malaysia, Indonesia, and Thailand that has been used for centuries as a remedy for a large number of ailments; the predominant alkaloid in kratom is mitragynine. A metabolite, 7-hydroxymitragynine, is formed naturally in leaf products and in the human body. Commercial formulations of 7-hydroxymitragynine are made from mitragynine using the identical chemical transformation, known as oxidation, that happens naturally. Mitragynine and 7-hydroxymitragynine (“7OH”) bind to the mu-opioid receptors but in a way that is fundamentally different than traditional opioids such as morphine.

Kratom and 7-Hydroxymitragynine products are used widely in the United States by people who seek autonomy over their healthcare, do not want the risk of overdose death from prescription opioids, nor the expense of prescription pharmaceuticals. Based upon manufacturer estimates, over 1.4 billion servings of 7OH have been dispensed in the US to as many as 3 million daily consumers; reasons for use include management of chronic pain; depression and anxiety; and as a remedy for opioid dependence; among other uses. The US Food and Drug Administration (FDA) has indicated that the US Drug Enforcement Administration (DEA) should consider scheduling 7OH and kratom as controlled substances.

Deaths from overdose.

Summary: There are no confirmed cases of fatalities from 7-OH alone despite years of commercial use.

America’s Poison Centers (formerly the National Poisoning Data System, or NPDS) have not reported any cases of fatalities from 7-hydroxymitragynine.¹ No cases of overdose death have been published in the peer-reviewed literature; there are no confirmed cases of fatalities from 7-OH alone despite years of commercial use. Furthermore, the 2026 abstracts at the Scientific Assembly of the American College of Medical Toxicology presented no overdose deaths. Nonetheless, two cases are notable:

A single case contained in the America’s Poison Centers database involved a hospice patient who suffered a blood clot in the lungs; this person had severe complications of treatment of the blood clot. Because this individual died six days after entering the hospital, acute 7OH toxicity could not have been the cause of death.

A young male with acute overdose was reported in Journal of Toxicology/Clinical Toxicology 2025. The patient, with a history of fentanyl dependence treated with Suboxone that he stopped because of prohibitive cost, had been fired from a menial job. After being fired, he drank excessive alcohol and was found not breathing and comatose by family members who called 911. Paramedics administered an extraordinary amount of naloxone (the antidote for opioid toxicity). He was transported to a hospital where he was admitted for a 72 hr suicide hold. The

¹ This is not because medical toxicologists are unaware of 7OH; on February 15, 2025, poison control centers added 7OH as a specific substance for data collection purposes. No deaths have been reported since that time. Moreover, poison center data does not show a spike in cases related to kratom since commercial sales of 7OH began in 2023. Even if 7OH use was misattributed as kratom use, a signal of overdose risk still does not exist.

clinicians never proved ingestion of 7OH nor disproved use of fentanyl despite the massive doses of naloxone needed for reversal; the publication did, however, demonstrate the presence of 7OH in commercial products found in his home.

Press releases describing 7OH overdose death generated by the Los Angeles County Medical Examiner Office lack scientific rigor. In none of the cases did the medical examiner measure the concentration of 7OH at autopsy.² Second, one-third of cases did not have an internal examination that may have identified an alternative cause of death. Third, none of the cases considered decedents' previous medical histories that may have contributed to death. Finally, none of these cases have been published in the peer-reviewed literature. Science by press release is not a typical approach to expanding the body of knowledge.

Overdose and toxicity: An increase in poison center calls reported in a 2026 Morbidity and Mortality Weekly Report conflated kratom and 7-hydroxymitragynine exposures—and incorrectly attributed the rise solely to commercial sales of 7OH, even though 7OH products had been in the marketplace for nearly two years prior to the dataset used in the publication.

Pediatric considerations: Pediatric exposure to 7OH is rare. Exposures are sporadic and wide geographic distribution. Some major centers (e.g., the children's hospital in Columbus, OH; Harvard's pediatric hospital, and major pediatric centers in Los Angeles have not seen a single pediatric case.

Evidence of outbreaks of multiple poisonings. No outbreaks of multiple poisonings have been reported. Calls to poison centers demonstrate that cases are sporadic and not tied to a geographical location as would be expected from severe toxicity that arises in multiple consumers. For comparison, substances such as fentanyl analogues, tianeptine, and isonitazenes produced clear surges in deaths and enforcement activity—conditions not currently seen with 7-hydroxymitragynine. In other words, substances such as fentanyl, tianeptine, and isonitazenes posed a clear and "imminent hazard" to public safety, indicating a rapidly escalating threat that is absent with 7OH.

Substance Use Disorder

Summary: The severity of substance use disorder from 7-hydroxymitragynine and the intensity of withdrawal may be less than that from comparable substances.

Laboratory research in animal models has demonstrated place preference for 7OH, a finding that suggests the possibility of abuse potential. Human data may be different. Emerging studies in a nationwide sample of drug treatment facilities performed at Johns Hopkins University has found that individuals seeking treatment for substance use disorder from 7OH is uncommon; in fact, among persons with substance use disorder from kratom-related products, 7OH is fourth on the list of substance classes despite months of commercial sales. In addition, the medical literature suggests that if buprenorphine is needed to ameliorate opioid withdrawal (and it is not always required), the

² The ratio of mitragynine to 7OH in forensic examinations is critically important. Kratom ingestion leads to formation of 7OH as a metabolite of mitragynine, but only in a small amount; in deaths following kratom use, therefore, the ratio of mitragynine:7OH would be very high, such as greater than 20 parts mitragynine to 1 part 7OH. Use of 7OH commercial products would lead to the reverse where a very high ratio of 7OH:mitragynine occurs.

dose needed to control 7OH withdrawal symptoms is less than that needed for treatment of withdrawal from classical opioids.

Pharmacology

Summary: 7-Hydroxymitragynine has pharmacology that limits its biological effects, toxicity, and overdose risk.

The basic science of 7OH partly explains the lack of public health signal of adverse effects from 7OH. Chiang (European Journal of Drug Metabolism and Pharmacokinetics, 2025) demonstrates that the first-pass metabolism (i.e., the amount of a dose that survives the liver) of orally administered 7OH is extremely low at 2.7-3%. Half of the surviving 7-hydroxymitragynine distributes into red blood cells and the remainder is subjected to additional degradation in the blood stream. Detoxifying molecules prevent entry of 7-hydroxymitragynine into the brain where it resides for only a brief time. 7-Hydroxymitragynine is rapidly eliminated from the body.

Even more importantly, federally funded research conducted at Memorial Sloan-Kettering Cancer Center identified that 7-hydroxymitragynine does not induce respiratory (DoD award number W81XWH-17-1-0256, “Safer nonaddicting and nonabusable analgesics....; Majumdar, PI).³

An explanation for the lack of respiratory depression is also found in basic pharmacology research. In 2021, Obeng (Journal of Pharmacology and Experimental Therapeutics, 2021) demonstrated that the binding-activity curve at the mu-opioid receptor for 7OH is nearly identical to that of buprenorphine, a molecule used in the treatment of opioid dependence and as an antidote for acute opioid poisoning. Thus, multiple lines of evidence funded by independent federal agencies or produced before the appearance of commercial 7OH products have indicated lack of respiratory effects from 7OH use.

Prepared by Edward W Boyer MD PhD

³ A separate study (Zuarth-Gonzales, Journal of Pharmacology and Experimental Therapeutics 2025) found 7OH could induce respiratory depression depending on study conditions. Several features of this research are noteworthy. First, 7OH was administered intravenously, a route never used by consumers. Second, the study used respiratory rate and minutes ventilation as a surrogate for respiratory depression, when the FDA-indicated measure for opioid respiratory effects is the ventilatory response to hypercapnia. Third, a commercial competitor to 7OH products has heavily funded that research group, a fact not revealed in conflict of interest statements.

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July 17, 2026

Jeffrey K. Shapiro
King & Spalding, LLP
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Washington, D.C. 20006

Dear Mr. Shapiro,

I am a medical toxicologist with 31 years of clinical experience. I am a Professor at UMass Chan Medical School and the director of our Division of Medical Toxicology (likely the largest such Division in the United States). In my clinical practice I have treated thousands of patients with overdoses of various substances. I have been asked by you to consider and opine upon the following topics related to 7-hydroxymitragynine (7-OH).

1. Whether the available human clinical data support Drug Enforcement Administration (DEA) publication of a temporary order scheduling an article containing greater than 1 mg of 7-OH in schedule I of the Controlled Substances Act.
2. Whether the DEA's proposed 1 mg-per-article threshold is scientifically justified in light of available human pharmacokinetic (PK), safety, and tolerability data.
3. Whether a higher threshold, including one informed by observed in vivo conversion of mitragynine to 7-OH, is scientifically more appropriate.
4. Whether the DEA's proposal for both a "greater than 0.050 percentage by weight/weight, weight/volume, or volume/volume" per article concentration standard and a "greater than 1.00 milligram" per article standard creates scientific and regulatory ambiguity, and whether the 0.050% standard should be deleted in favor of a single threshold expressed in milligrams per article.
5. Any assumptions, methodological limitations, or safety caveats that should accompany your opinion.

To accomplish this, I performed literature searches and reviews in PubMed and Google Scholar; reviewed relevant scientific articles; and reviewed relevant citations from those identified scientific articles. I then synthesized these various informative sources to answer the above questions.

Before answering the above-listed five questions for you, it is worth considering the critical medical and toxicological questions at issue. To me, those critical questions are:

1. As 7-OH is formed from in vivo conversion of mitragynine (MTG), are there PK data regarding the rapidity and maximum concentration of 7-OH after mitragynine use?
2. Are there PK data regarding 7-OH when used alone (that is, without mitragynine)?
3. Can we estimate what a serum concentration of 7-OH would be under various assumptions?
4. Has a serum/blood concentration for 7-OH been identified at or above which a health risk (namely, respiratory depression) may occur? That is, is there an established pharmacodynamic effect at a defined concentration level?

I will begin by discussing these four medical/toxicological points.

I. Pharmacokinetic data of in vivo 7-OH formation from mitragynine

The serum/blood concentrations and duration of 7-OH are dictated by the rate at which MTG is metabolized. In humans, MTG is converted into 7-OH primarily through CYP3A4-mediated metabolism in the liver. While CYP3A4 is the major driver, enzymes such as CYP2C19 and CYP2D6 contribute to this conversion to a lesser extent.

Both the formation and clearance of 7-OH are CYP3A-dependent. Ketoconazole (a CYP3A inhibitor) increased systemic exposure to 7-OH by 130% in rats and increased its pharmacodynamic potency 4- to 10-fold, confirming that CYP3A is the principal clearance pathway for 7-OH itself, not just its formation from MTG. In humans, women generally exhibit higher CYP3A4 activity, which would lead to *lower* 7-OH concentrations compared to men. 7-OH undergoes further oxidative metabolism to at least two identified products:

- Mitragynine pseudoindoxyl — formed via a CYP3A-mediated skeletal rearrangement. This metabolite is a multifunctional μ -agonist/ δ - κ antagonist with >2-fold μ -opioid receptor-biased agonism relative to its precursors, and it exhibits reinforcing properties and gastrointestinal (GI) transit inhibition (Chakraborty 2021; Avula 2026).
- 3-Dehydromitragynine — formed through a non-CYP oxidation pathway. This metabolite is also generated when 7-OH degrades under simulated gastric conditions (up to 27% degradation in simulated gastric fluid) (Manda 2014; Chakraborty 2021; Avula 2026).

The most detailed human data regarding 7-OH come from the study by Huestis *et al.* (Huestis MA. *Ther Drug Monit* 2026;00:1–11), which characterized plasma PK of both mitragynine and 7-OH-MTG after oral kratom extract (39.5% mitragynine) at escalating doses.

- After single doses of 9.9, 29.6, and 59.2 mg mitragynine, mean 7-OH C_{max} values were 6.8, 13.7, and 30.5 ng/mL, respectively, with a median T_{max} of 1.3–1.7 h.
- After 15 daily doses at the same escalating levels, mean 7-OH C_{max} values were 5.4, 16.6, and 33.9 ng/mL with T_{max} of 1.7–2.3 hours.
- C_{max} and AUC increased approximately dose-proportionally (less than 1.4 x greater than expected based on strict dose proportionality).
- Higher 7-OH to mitragynine ratios were observed after multiple doses compared to single doses, and at lower compared to higher doses, suggesting possible CYP3A enzyme saturation at higher doses.

Because 7-OH formation is rate-limited, its elimination half-life when formed as a metabolite is parallel to that of MTG, approximately 4.7 to 24.7 hours in human subjects depending on the study and dose.

II. Pharmacokinetic data regarding 7-OH when used alone

After exhaustive literature review, I could find no data regarding the PK (or pharmacodynamics) of 7-OH when ingested alone in humans. In their absence, I have used fundamental PK principles to estimate human 7-OH concentrations.

Several factors suggest that absolute oral bioavailability of 7-OH is likely modest:

- Gastric instability: 7-OH degrades up to 27% in simulated gastric fluid, with approximately 23% converting back to mitragynine and additional degradation to 3-dehydromitragynine. Only 6% degradation occurs in simulated intestinal fluid (Manda 2014).
- Rapid hepatic metabolism: 7-OH has a microsomal half-life of only 24 minutes in human liver microsomes, indicating extensive first-pass metabolism (Manda 2014).
- CYP3A-dependent clearance.

PK studies in rodents have been performed. Chiang *et al.* (Chiang 2025), performed oral and IV PK studies of 7-OH in male rats. The volume of distribution was 2.7 ± 0.4 L/kg and the clearance was 4.0 ± 0.3 L/h/kg after IV administration. Notably, the volume of distribution of MTG in humans and rodents show 97% agreement. It is expected that the volume of distribution in humans would also be low. After 5 mg/kg oral dosing, a $C_{\max}$ of 28.5 ± 5.0 ng/ml and $T_{\max}$ of 0.3 ± 0.1 h were observed. Mitragynine pseudoindoxyl (MGPI), the metabolite of 7-OH that is a more potent μ -opioid agonist than 7-OH, had a parent-to-metabolite ratio for MGPI following IV 7-OH of $0.5 \pm 0.1\%$, indicating very limited systemic exposure to MGPI.

Further PK data from rats are also informative. 7-OH exhibits >90% plasma protein binding by equilibrium dialysis, similar to mitragynine (Manda 2014). Such high binding limits the free fraction available for tissue distribution and receptor engagement. Furthermore, Yusof *et al.* (Yusof 2019) determined that 7-OH has lower blood-brain-barrier (BBB) permeability than mitragynine (that is, 7-OH does not as readily cross into the brain); 7-OH shows a tendency toward P-gp-mediated efflux (efflux ratio 1.39); and less than 10% of unbound 7-OH in plasma crosses the BBB ($K_p,uu < 0.1$), indicating extensive efflux.

So, despite the limitations of data in humans, all available evidence supports my conclusions.

III. Estimating the serum concentration of 7-OH under various assumptions

Another way to look at the 7-OH PK issue is by taking a hypothetical 1 mg dose of 7-OH and calculating its apparent serum concentration given PK principles.

For an adult male (averaging 70 to 75 mL of blood per kg): An 80 kg male has a total blood volume of roughly 5,600 to 6,000 mL.

- At 6,000 mL: $1 \text{ mg} / 6,000 \text{ mL} \approx 0.000167 \text{ mg/mL}$ (167 ng/mL)

For an adult female (averaging 65 mL of blood per kg): A 70 kg female has a total blood volume of roughly 4,550 mL.

- At 4,550 mL: $1 \text{ mg} / 4,550 \text{ mL} \approx 0.000220 \text{ mg/mL}$ (220 ng/mL)

However, the distribution of 7-OH out of the bloodstream and into extravascular tissues (fat, muscle, etc) must be considered. The actual concentration in the blood is dictated by the drug's apparent Volume of Distribution (V_d), using this formula:

$$C = \frac{\text{Dose}}{V_d \times \text{Weight}}$$

If the rat V_d of 7-OH, ($V_d = 2.7$ L/kg; NB: the rat and human volumes of distribution for MTG agree

with each other by 97%) is used, then the calculation is as follows:

- An 80 kg person has a distribution volume of 216 L (216,000 mL).
- Blood concentration: 1 mg / 216,000 mL $\approx$ 0.0000046 mg/mL (4.6 ng/mL)

This 4.6 ng/mL assumes instantaneous and complete absorption of the 1 mg 7-OH dose.

Protein binding of 7-OH also must be considered. Studies by Manda and Obeng (Obeng 2019) have demonstrated the 7-OH and related MTG metabolites are extensively protein bound (~90%). However, only the *unbound* fraction is free to interact with receptors and cross cellular membranes.

$$C_{free} = C_{total} \times f_u$$
$$C_{free} = 4.63 \text{ ng/mL} \times 0.10 = 0.463 \text{ ng/mL}$$

Thus, in under these assumptions, a single 1 mg dose of 7-OH would result in an approximate serum concentration of just **0.46 ng/mL**, well below the measured serum concentrations in studies by Huestis. By extension, a 10 mg dose of 7-OH would yield an estimated concentration of 4.6 ng/mL (again, roughly 7 times lower than the 7-OH concentration measured in Huestis' highest dose).

IV. The 7-OH serum/blood concentration at or above which a health risk (namely, respiratory depression) may occur

Given 7-OH's ability to bind to the μ -opiate receptor, the greatest concern regarding overdose of 7-OH is respiratory depression.

There is no scientifically established serum/blood concentration of 7-OH at or above which respiratory depression is likely to occur. Aside from the lack of clinical data, the most compelling reason for this lack of respiratory effect threshold is that 7-OH exhibits atypical opioid pharmacology. 7-OH is a potent *partial agonist* at the μ -opioid receptor. Unlike full agonists (such as morphine or fentanyl), partial agonists often demonstrate a "ceiling effect" for respiratory depression in preclinical models. While 7-OH has a much higher binding affinity and potency at the μ -opioid receptor than mitragynine, its G-protein-biased signaling complicates setting a linear concentration-to-risk threshold. These facts (and others), make the likelihood of respiratory depression *less likely* with 7-OH.

Given this lack of clinical data on respiratory effects, perhaps the best and most voluminous data on the safety of 7-OH comes from a review of Poison Center and other data (e.g., DEA) and the manufacturers of 7-OH.

Reports of 7-OH toxicity and deaths potentially linked to 7-OH in the realm of medicine and medical toxicology are generally rare. However, the real-world use of 7-OH products (in which the most common 7-OH dose sold in the U.S. is the 50 mg tablet) is tremendous, with more than 93,381,000 tablets sold in the U.S. (which is likely an underestimate of the true consumption of 7-OH). Reconciling the large use of 7-OH products with the relative dearth of reports of 7-OH toxicity resulting in significant adverse health effects can only lead to the conclusion that 7-OH is relatively safe, even at large doses such as 50 mg or more.

While it is hard to prove a negative, I believe that data related to total production of 7-OH products

provides a compelling argument for 7-OH's safety when used at reasonable doses.

V. My answers to the questions posed to me

- Q1. Whether the available human clinical data support DEA publication of a temporary order scheduling an article containing greater than 1 mg of 7-OH in schedule I of the Controlled Substances Act.

Available human and animal studies *do not* support a 7-OH threshold/dose that low. Available human and animal studies support establishment of a 7-OH threshold/dose higher than 1 mg per article. Evidence for this opinion is found in the human PK data by Huestis *et al.* which showed that 7-OH C_{max} from mitragynine doses up to 59.2 mg were well-tolerated. Additionally, we have real-world use of millions of 50 mg tablets without proportionate adverse events. It is also worth mentioning that I could find no rationale or justification for DEA's 1 mg limit recommendation.

- Q2. Whether the DEA's proposed 1 mg-per-article threshold is scientifically justified in light of available human PK, safety, and tolerability data.

It is my opinion that the 1 mg-per-article threshold *is* overly conservative based upon pre-clinical and clinical science. Such a limit also lacks validity given what is known about 7-OH use in the United States.

- Q3. Whether a higher threshold, including one informed by observed in vivo conversion of mitragynine to 7-OH, is scientifically more appropriate.

It is my opinion that a 5 mg per article dose is safe and does not present an imminent hazard to the public safety. Support for this opinion is found in the Huestis human PK studies, in which a high 7-OH C_{max} of 30.5 ng/mL was found after mitragynine dosing. Furthermore, the partial agonist/ceiling effect of 7-OH lends another pharmacodynamic level of safety to the chemical. In fact, given all of the data that I have reviewed, it would also be my opinion that a 10 mg dose is safe.

- Q4. Whether the DEA's proposal for both a "greater than 0.050 percentage by weight/weight, weight/volume, or volume/volume" per article concentration standard and a "greater than 1.00 milligram" per article standard creates scientific and regulatory ambiguity, and whether the 0.050% standard should be deleted in favor of a single threshold expressed in milligrams per article.

It is my opinion that the DEA's use of both a 0.050% concentration standard and a 1 mg-per-article standard is not necessary and is unduly complicated. It is not the concentration (weight/weight) that is important, but rather the milligrams of 7-OH per dose. Requiring just a milligram-based standard allows for much clearer dosing guidance across products of varying size and formulation.

- Q5. Any assumptions, methodological limitations, or safety caveats that should accompany your opinion.

My opinions are based upon available data. Ideally, PK studies of 7-OH would be performed. Such studies would eliminate some of the extrapolations being relied upon for my opinion (e.g., 7-OH metabolite data from Huestis; rodent PK data; and protein binding estimates). Due to the lack of human PK data for 7-OH, the most appropriate safe dosage is deemed to be 5 mg per

article.

A handwritten signature in black ink, appearing to read 'S. Bird', with a stylized, sweeping flourish extending from the end.

Steven B. Bird, MD FACMT FACEP

FOR IMMEDIATE RELEASE

August 21, 2025

Contact: media@hartsupporter.com

LEADING RESEARCHERS REJECT FDA’S POSITION, FIND NO EVIDENCE OF HARM FROM 7-OH IN RESPONSE FROM SHAMAN BOTANICALS TO FDA

[WASHINGTON, DC] Shaman Botanicals has issued a formal response to the FDA’s recent warning letter on its 7-Hydroxymitragynine (7-OH) products, **backed by reports from leading researchers at Johns Hopkins, Harvard, UCLA, and other institutions**. The company argues that current scientific evidence supports the safety of 7-OH, directly countering FDA’s claims. Experts consulted—including those previously cited by FDA—stress that 7-OH should not be considered a public health crisis and that available data shows no evidence of overdose deaths, respiratory depression, or widespread dependence.

The experts also emphasize that **7-OH is a naturally occurring kratom-derived compound**, not synthetic, and that **preclinical studies commissioned by Shaman found no serious toxicity**. Collectively, the findings point toward a favorable safety profile and call for continued study rather than scheduling. Shaman Botanicals maintains its commitment to scientific transparency and consumer safety, urging FDA to follow the science.

Key Points:

1. Experts previously cited by the FDA now disagree with its recommendation to schedule 7-OH. They argue the agency’s position is not supported by the latest scientific evidence.
2. **No deaths or confirmed cases of respiratory depression have been linked to 7-OH.** Researchers claim this is significant given millions of adults are estimated to have tried it.
3. **Researchers say that 7-OH has “no unreasonable risk of overdose deaths”** based on reviews of nationwide databases. Poison control and toxicology centers across the U.S. confirm no evidence of widespread harm.
4. **Scientists confirm that 7-OH is a natural metabolite of kratom**, not a synthetic substance. It is produced both by the plant itself and through metabolism in the human body.

See the full Shaman Botanicals, LLC Response Below:

Shaman Botanicals, LLC Responds to FDA Warning Letter with Scientific Reports by Renowned Experts

Shaman Botanicals, LLC, a leading innovator in botanical wellness, has formally responded to a recent warning letter from the U.S. Food and Drug Administration (FDA) regarding its 7-Hydroxymitragynine (7-OH) products. In support of its mission and to defend its products, attorneys for Shaman Botanicals have assembled an interdisciplinary team of renowned researchers, scientists, and clinicians to prepare reports that directly address the FDA’s claims concerning 7-OH.

"The FDA asserts that there is insufficient information about the safety of our 7-OH products. However, the scientists consulted by our legal team have found that the evidence to date supports the safety of our products," stated Stephen "Vince" Sanders, II, founder of Shaman Botanicals.

Scientists Cited by the FDA Express Concerns About the Scheduling of 7-OH

Scientific reviews were provided by **Dr. Kirsten Smith, PhD**, widely regarded as the leading clinical researcher on Kratom in the United States and an Assistant Professor at Johns Hopkins University, and **Dr. Edward Boyer, MD, PhD**, a Harvard-educated medical toxicologist with extensive experience in Kratom research.

The support for further study and regulation of 7-OH products by Drs. Smith and Boyer reflects a growing and dynamic shift in the scientific community's perspective on 7-OH. Work by Drs. Smith and Boyer was cited by the FDA in their recent report recommending the scheduling of 7-OH. Both Drs. Smith and Boyer are currently leading projects to better understand 7-OH use patterns in the United States. But, having looked at the latest data, they do not agree with FDA's position.

Other scientists cited in Shaman's response to the FDA include **Dr. Michael Levine, MD**, a medical toxicologist educated at Chicago Medical School and an associate professor at the University of California-Los Angeles, recognized for his research in botanical toxicology, and **Dr. Jakob Moskovitz, D.Sc.**, a Technion-Israel Institute of Technology-trained biochemist and internationally recognized expert in molecular biology. Drs. Levine and Moskovitz were provided with three preclinical studies commissioned by Shaman and asked to comment on them. Shaman's response also cited **Dr. Glenn Prestwich, PhD**, a Stanford-educated chemist and internationally recognized authority on natural products chemistry and chemical biology, and **Dr. Andrew Monte, MD, PhD**, the Associate Director of the Rocky Mountain Poison & Drug Center in Denver, Colorado, and an expert in nationwide monitoring for adverse events from opioids.

7-OH Should Not be Considered a Public Health Crisis

Dr. Smith strongly disagrees with the FDA's recent recommendation to schedule 7-OH. According to Dr. Smith, the use of 7-OH does not constitute a public health crisis. Furthermore, Dr. Smith asserts that, much like kratom, 7-OH currently offers a net benefit to its users.

According to Dr. Smith, who estimates that at least 4 million adults have tried 7-OH, there are no confirmed human cases of respiratory depression caused by 7-OH. Additionally, reports of physical dependence or substance use disorder related to 7-OH are even rarer than those associated with kratom.

7-OH Is Not Causing Widespread Illness or Injury

Dr. Boyer reviewed nationwide toxicology databases, including poison center data, and found limited evidence of acute overdose risk from 7-hydroxymitragynine. No deaths have been attributed to 7-hydroxymitragynine, a finding confirmed by America's Poison Center in their August 12, 2025 report.

As Dr. Boyer stated, “A signal arising from overdose death from 7-hydroxymitragynine is absent. Based upon data available at this time, given the degree of population exposure, the lack of findings suggests that 7-hydroxymitragynine carries no unreasonable risk of overdose death.”

Dr. Monte arrived at the same conclusion as Dr. Boyer, that there is currently no evidence of a safety signal regarding 7-OH as reported by independent poison control and toxicology centers across the United States.

Dr. Boyer also disagreed with the FDA’s report regarding the putative safety or danger of 7-OH. According to Dr. Boyer, the FDA assessment relies on animal studies receiving intravenous 7-hydroxymitragynine (a route of administration never used in humans) and overlooks existing data indicating low bioavailability, rapid clearance, and poor brain penetration characteristic of oral 7-hydroxymitragynine.

7-OH is a Natural Product; Not Synthetic

Opponents of 7-OH often mistakenly label these kratom-derived products as “synthetic.” According to Dr. Prestwich, “7-OH is not manufactured synthetically.” Instead, 7-OH is a naturally occurring selective oxidation product of mitragynine, already present among the indole alkaloids in kratom plant extract. These products contain 7-OH that is chemically identical to the natural metabolite of mitragynine produced by the kratom plant and through metabolism in the human body. Thus, 7-OH is a natural product.

Pre-Clinical Data Conducted by Shaman Botanicals Supports the Favorable Safety Profile of 7-OH

Drs. Levine and Moskovitz each reviewed three preclinical studies commissioned by Shaman: (1) “Acute Toxicity Assessment of 7-Hydroxymitragynine in Rodents,” (2) “Pilot Safety Study of Escalating Doses of 7-Hydroxymitragynine and Mitragynine Pseudoindoxyl Administered Daily for 7-day Intervals to Mature Dogs,” and (3) “An In-Vitro Pre-Clinical Safety Comparison of Mitragynine and its Oxidative Metabolites: 7-Hydroxymitragynine and Mitragynine Pseudoindoxyl.” In the two animal studies, no serious clinical toxicity or adverse events were observed.

Dr. Levine stated these studies support initial safety and warrant further evaluation. Dr. Moskovitz concluded, “Based on all the obtained data (i.e., in vitro studies, mice and dogs’ models), it seems that the therapeutic potency and safety profile of 7HMG are either the same or better compared to mitragynine.”

Current Science Supports a Reasonable Basis of Safety and Recommends Continued Study, Not Scheduling

Full copies of each expert report have been provided to the FDA. “The FDA pledged to follow the science. The aforementioned experts, some of the top scientists in the world, maintain the science supports a reasonable basis of safety in our 7-OH products” Sanders said.

Commitment to Transparency and Consumer Safety

Shaman Botanicals remains deeply committed to transparency, scientific rigor, and consumer welfare. The company welcomes continued dialogue.